

RISTNA, Estonia

WMO: 261150

Lat: 58.9209N

Lon: 22.0664E

Elev: 27

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.5	13.3	3.0	6.4	10.5	7.4	8.1	15.4	28.3	37.3	25.1	36.2	4.9	90	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	77.5	67.6	74.1	65.9	71.1	64.2	69.1	75.1	67.1	72.5	65.4	70.0	5.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.7	72.5	64.9	92.4	70.4	63.3	87.1	68.5	33.2	75.4	31.6	72.7	30.3	70.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	21.3	17.9	DB	4.6	80.9	7.6	4.0	-0.8	83.7	-5.2	86.1	-9.5	88.3	-15.0	91.2	(4)
(5)			WB	4.1	70.8	7.3	2.9	-1.2	72.9	-5.4	74.6	-9.5	76.2	-14.8	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	29.8	28.4	31.9	39.2	47.9	56.1	63.9	63.0	55.9	46.9	39.5	34.5
	DBStd	13.62	7.84	7.93	5.15	5.34	6.38	5.21	4.89	4.36	5.12	5.93	6.29	7.22
	HDD50	3183	627	606	561	330	116	6	0	0	13	129	315	480
	HDD65	7440	1092	1026	1026	775	531	273	81	90	274	562	765	945
	CDD50	1300	0	0	0	5	51	189	430	403	190	32	0	0
	CDD65	82	0	0	0	0	1	6	46	28	1	0	0	0
	CDH74	291	0	0	0	0	2	36	181	72	0	0	0	0
	CDH80	29	0	0	0	0	0	2	18	9	0	0	0	0
Wind	WSAvg	7.5	8.4	7.7	6.9	6.3	6.0	6.4	6.3	7.5	7.6	8.3	8.3	9.7
Precipitation	PrecAvg	24.3	2.0	1.6	1.3	1.2	1.3	2.0	2.2	2.9	2.3	2.6	2.8	2.2
	PrecMax	33.3	5.3	3.6	2.6	2.4	2.2	4.4	4.4	5.8	5.4	4.9	5.1	6.3
	PrecMin	9.4	0.5	0.2	0.1	0.2	0.2	0.4	0.0	0.2	0.2	0.3	0.2	0.2
	PrecStd	4.8	1.0	0.8	0.6	0.7	0.6	1.0	1.2	1.5	1.2	1.3	1.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	40.5	46.6	62.1	72.9	78.7	82.6	80.3	69.1	59.9	51.3	47.6
		MCWB	42.2	38.0	41.1	51.8	60.0	65.9	70.3	69.6	62.0	56.5	49.7	46.0
	2%	DB	41.6	39.0	41.3	55.8	67.5	71.8	78.6	75.4	66.4	58.0	50.1	46.4
		MCWB	40.2	37.4	38.0	47.6	56.8	62.0	68.3	66.6	61.1	55.2	48.4	44.9
	5%	DB	40.1	38.0	39.4	51.0	62.3	67.5	75.5	72.5	64.4	56.5	48.7	44.9
		MCWB	38.5	36.7	37.0	44.8	54.0	59.6	67.3	65.4	59.9	54.0	47.0	43.2
	10%	DB	38.7	37.0	38.1	47.3	58.1	64.2	72.5	70.0	62.8	54.8	47.3	42.9
		MCWB	37.2	35.9	36.1	42.5	51.5	58.5	65.7	64.1	59.0	52.5	45.5	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	38.9	41.3	52.3	62.2	67.1	72.4	70.4	64.0	57.5	50.1	46.6
		MCDB	43.1	39.6	45.4	60.5	69.1	76.6	79.1	78.0	67.0	59.0	50.8	47.2
	2%	WB	40.6	37.7	38.6	48.4	58.0	63.6	70.2	68.1	62.1	55.7	48.7	45.2
		MCDB	41.4	38.5	40.5	53.6	65.3	69.9	76.1	73.7	64.9	57.3	49.7	46.2
	5%	WB	38.7	36.9	37.5	45.6	55.1	61.0	68.3	66.4	60.8	54.4	47.4	43.2
		MCDB	39.6	37.7	39.0	50.5	61.6	65.7	74.0	70.8	63.5	56.1	48.5	44.4
	10%	WB	37.4	36.0	36.4	43.0	52.0	59.0	66.5	64.9	59.5	52.9	45.7	41.5
		MCDB	38.5	36.8	37.7	46.4	57.1	63.3	71.7	68.8	62.1	54.6	47.0	42.7
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.2	6.9	9.3	10.6	9.7	10.0	9.2	7.9	6.3	5.2	5.4
		MCDBR	4.8	4.9	8.1	16.2	18.0	16.8	15.1	13.2	9.8	5.9	4.6	4.3
		MCWBR	5.0	4.6	6.1	9.6	9.9	8.8	8.0	7.3	6.9	5.1	5.2	4.6
	5% WB	MCDBR	4.9	4.8	7.1	14.6	15.6	13.8	13.2	10.6	7.8	5.3	4.4	4.1
		MCWBR	5.3	4.7	5.7	9.2	9.5	8.3	7.7	6.9	6.2	5.0	5.3	4.7
Clear-Sky Solar Irradiance	taub	0.291	0.320	0.339	0.374	0.368	0.364	0.393	0.385	0.355	0.339	0.322	0.281	
	taud	2.205	2.270	2.328	2.334	2.428	2.477	2.419	2.424	2.491	2.408	2.231	2.136	
	Ebn at Noon	163	220	253	263	274	277	265	258	248	213	148	121	
	Edh at Noon	15	23	28	33	33	32	33	30	24	20	14	11	
All-Sky Solar Radiation	RadAvg	86	259	750	1302	1820	1945	1824	1433	893	391	111	57	
	RadStd	13	46	79	140	165	106	141	124	104	50	23	8	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page